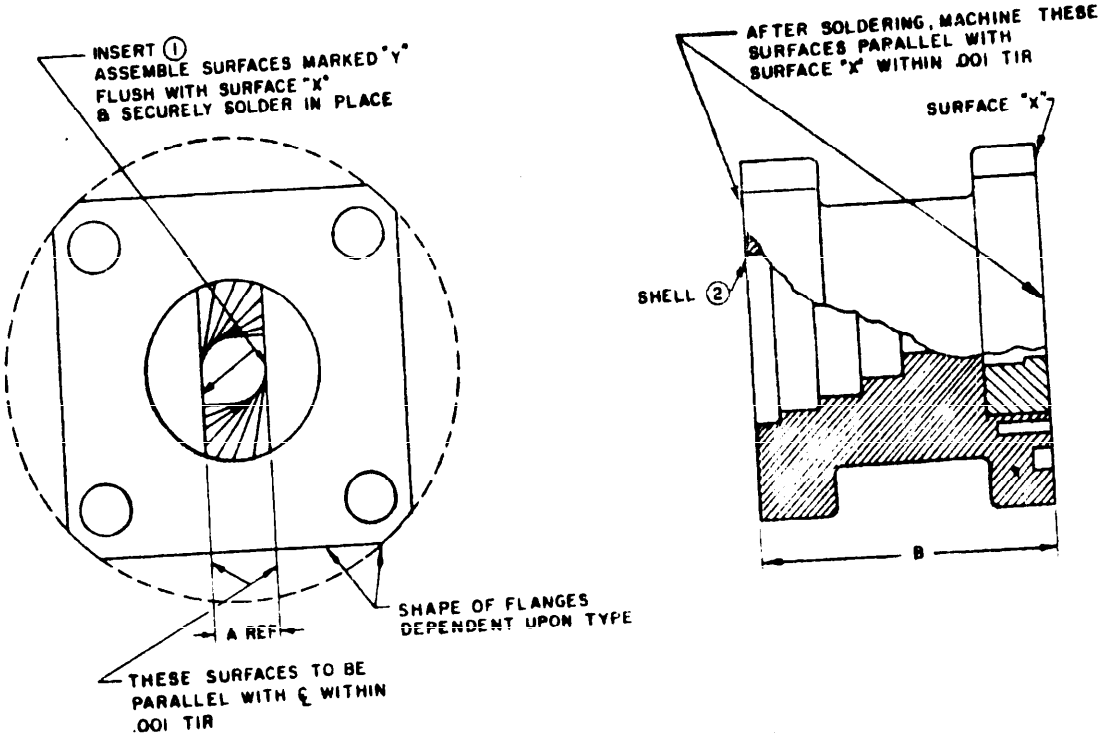
MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, 90-DEGREE STEP TWIST, CG-1134/U THROUGH CG-1147/U

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the issue in effect of Specification MIL-W-3970.



ASSEMBLY

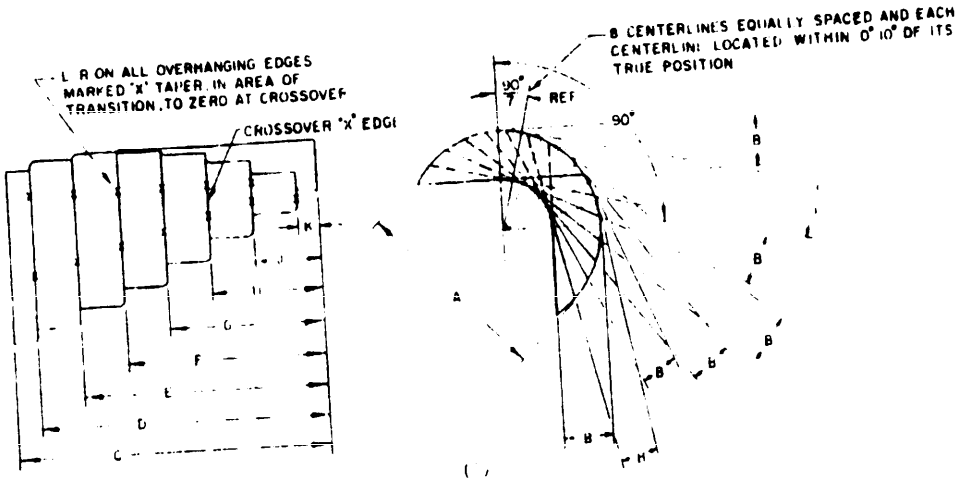
FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:05Z
Check the source to verify that this is the current version before use.

Military part no.	Assembly type	Item	Description	Material	Waveguide sections per MIL-W-85 from	Flanges equivalent to		Dimensions	
						Choke	Cover	A	B
M3970 4-001	CG-1134 U	1 2	Insert Shell	Brass	RG-53 U	UG-596A U per MIL-F-3922 19	UG-595 U per MIL-F-3922 18	0.170	0.729
M3970 4-002	CG-1135 U	1 2	Insert Shell	Aluminum	RG-121 U	UG-598A U per MIL-F-3922 21	UG-597 U per MIL-F-3922 20		
M3970 4-003	CG-1136/U	1 2	Insert Shell	Brass	RG-51 U	UG-600A U per MIL-F-3922 23	UG-51 U per MIL-F-3922 2	0.497	1.953
M3970 4-004	CG-1137 U	1 2	Insert Shell	Aluminum	RG-68 U	UG-137B U per MIL-F-3922 28	UG-138 U per MIL-F-3922 5		
M3970 4-005	CG-1138 U	1 2	Insert Shell	Brass	RG-52 U	UG-40B U per MIL-F-3922 24	UG-39 U per MIL-F-3922 1	0.400	1.566
M3970/4-006	CG-1139 U	1 2	Insert Shell	Aluminum	RG-67 U	UG-136B U per MIL-F-3922 27	UG-135 U per MIL-F-3922 4		
M3970/4-007	CG-1140/U	1 2	Insert Shell	Brass	RG-50 U	UG-343 U per MIL-F-3922 30	UG-344 U per MIL-F-3922 6	0.622	2.384
M3970/4-008	CG-1141/U	1 2	Insert Shell	Aluminum	RG-106 U	UG-440B U per MIL-F-3922 36	UG-441 U per MIL-F-3922 11		
M3970/4-009	CG-1142/U	1 2	Insert Shell	Brass	RG-48 U	UG-54B U per MIL-F-3922 26	UG-53 U per MIL-F-3922 3	1.340	4.936
M3970/4-010	CG-1143/U	1 2	Insert Shell	Aluminum	RG-75 U	UG-585A U per MIL-F-3922 17	UG-584 U per MIL-F-3922 16		
M3970 4-011	CG-1144/U	1 2	Insert Shell	Brass	RG-49 U	UG-148C U per MIL-F-3922 37	UG-149A U per MIL-F-3922 29	0.872	3.255
M3970/4-012	CG-1145/U	1 2	Insert Shell	Aluminum	RG-95 U	UG-406B/U per MIL-F-3922 31	UG-407 U per MIL-F-3922 12		
M3970/4-013	CG-1146 U	1 2	Insert Shell	Brass	RG-96 U	UG-600A U per MIL-F-3922 23	UG-599 U per MIL-F-3922 22	0.140	0.485
M3970 4-014	CG-1147/U	1 2	Insert Shell	Brass	RG-91 U	UG-541A U per MIL-F-3922 13	UG-419 U per MIL-F-3922 10	0.311	1.079

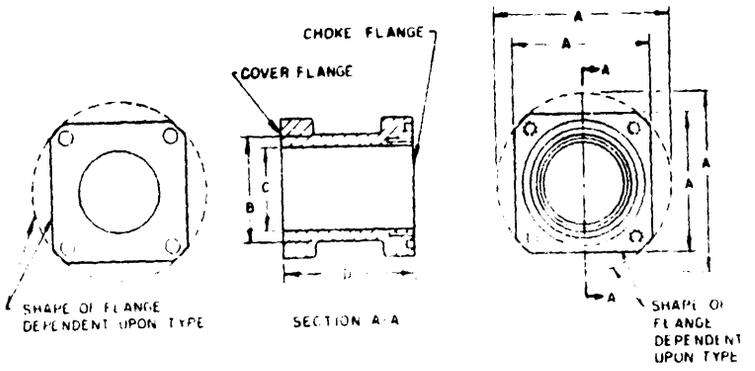
1. All dimensions in inches.
2. Unless otherwise specified, tolerances are 1 64 on fractions and 0.005 on decimals.
3. Hardware shall be furnished with choke flanges.

Source: https://assist.dla.mil -- Downloaded: 2026/0-17T03:05Z
Check the source to verify that this is the current version before use.



Assembly type	Dimensions										
	A	B	C	D	E	F	G	H	J	K	L
CG-1134/U CG-1135/U	0.4484 -0.0006	0.139 +0.001	0.745	0.683 0.003	0.578 0.003	0.472 0.003	0.367 0.003	0.262 0.003	0.156 0.003	0.051 0.003	0.014
CG-1136/U CG-1137/U	1.1990 -0.0009	0.351 +0.001	1.968	1.830	1.548	1.266	0.984	0.702	0.420	0.138	0.030
CG-1138/U CG-1139/U	0.9621 -0.0008	0.281 +0.001	1.581	1.467	1.241	1.015	0.789	0.563	0.337	0.111	0.030
CG-1140/U CG-1141/U	1.4658 -0.0009	0.422 +0.001	2.399	2.234	1.889	1.545	1.200	0.856	0.511	0.167	0.044
CG-1142/U CG-1143/U	3.0351 -0.0012	0.648 +0.002	4.969	4.625	3.912	3.199	2.486	1.773	1.060	0.347	0.094
CG-1144/U CG-1145/U	2.0006 -0.0010	0.565 +0.001	3.270	3.050	2.580	2.110	1.640	1.170	0.700	0.230	0.060
CG-1146/U	0.2986 -0.0005	0.079 +0.001	0.500	0.454 0.002	0.384 0.002	0.314 0.002	0.244 0.002	0.174 0.002	0.104 0.002	0.034 0.002	0.010
CG-1147/U	0.6633 -0.0007	0.176 +0.001	1.094	1.011	0.855	0.699	0.543	0.387	0.231	0.075	0.020

Source: https://assist.dla.mil -- Downloaded: 202610-17T03:05Z
Check the source to verify that this is the current version before use.



Assembly type	Dimensions				
	A		B	C	D
	Square	Round			
CG-1134/U CG-1135/U	7/8	---	0.635	0.4490 +0.006	0.746
CG-1136/U CG-1137/U	1-7/8	---	1-25/64	1.2000 -0.0009	1.968
CG-1138/U CG-1139/U	1-5/8	---	1-1/8	0.9630 +0.0008	1.581
CG-1140/U CG-1141/U	---	3-1/8	1-23/32	1.4670 -0.0009	2.399
CG-1142/U CG-1143/U	---	5-5/16	3-1/2	3.0370 -0.0012	4.969
CG-1144/U CG-1145/U	---	3-5/8	2-5/16	2.0020 +0.0010	3.270
CG-1146/U	3/4	---	0.424	0.2990+0.0005	0.500
CG-1147/U	1-5/16	---	25/32	0.6640+0.0007	1.094

Solder. All parts shall be securely soldered in place using solder conforming to composition Sn60 of QQ-S-571.

Finish. After machining, brass shall be finished in accordance with finish No. M351, type II of MIL-F-14072 and a rhodium or palladium flash shall be applied.

Engineering information. Waveguide assemblies covered by this specification sheet exhibit a maximum VSWR of less than 1.05:1 over 40 percent of the usable bandwidth of the assembly. The following electrical characteristics are furnished for information only:

Source: https://assist.dla.mil -- Downloaded: 202640-17T03:05Z
Check the source to verify that this is the current version before use.

Assembly type	Frequency range kmc/sec-TE_{10} mode	Power rating at sea level in kw	Used with rectangular waveguides per MIL-W-85
CG-1134/U	18.00 - 26.50	20	RG-53/U
CG-1135/U	18.00 - 26.50	20	RG-121/U
CG-1136/U	7.05 - 10.00	175	RG-51/U
CG-1137/U	7.05 - 10.00	175	RG-68/U
CG-1138/U	8.20 - 12.40	100	RG-52/U
CG-1139/U	8.20 - 12.40	100	RG-67/U
CG-1140/U	5.85 - 8.20	280	RG-50/U
CG-1141/U	5.85 - 8.20	280	RG-106/U
CG-1142/U	2.60 - 3.95	1,100	RG-48/U
CG-1143/U	2.60 - 3.95	1,100	RG-75 U
CG-1144/U	3.95 - 5.85	700	RG-49/U
CG-1145/U	3.95 - 5.85	700	RG-95/U
CG-1146/U	26.50 - 40.00	11	RG-96/U
CG-1147/U	12.40 - 18.00	60	RG-91 U

Custodians:
Army - EL
Navy - Ships
Air Force - ASD

Preparing activity:
Navy - Ships
(Proj 5985-0493)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:05Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:05Z
Check the source to verify that this is the current version before use.